Math 118: Mathematical Methods of Data Theory

Lecture 22: PageRank

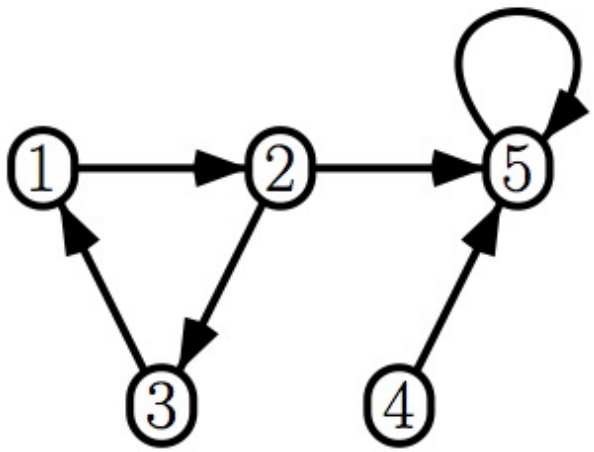
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March 4, 2020

Overview

- Let $G = (V, E)$ be a **directed graph**.
- Edges $(i, j) \in E$ are ordered: $(i, j) \neq (j, i)$.
- **Examples:**
  - Twitter: vertices are users, edges are follows.
  - Gene regulatory networks: vertices are genes, edges represent activation.
  - World Wide Web: vertices are web pages, edges are hyperlinks.
- **Convention:** edge (i, j) points from i to j. Call i the tail and j the head.



$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

- $d_i^{\text{out}} = \sum_j A_{ij}$ = number of outlinks from i. *e.g.* $d_2^{\text{out}} = 2$
- $d_i^{\text{in}} = \sum_j A_{ji}$ number of inlinks to i. *e.g.* $d_2^{\text{in}} = 1$

PageRank

- **Motivation:** Want to rank all websites relating to a particular topic (*e.g.* "UCLA") using only link structure.
- **Idea:** Good websites should be linked to by many other websites.
- **Improved Idea:** Good websites should be linked to by many other **good** websites.
- Let r_i = score of i-th website. Should be related to scores of all j linking to i:

$$r_i = \sum_{j:\ (j,i)\in E} \frac{r_j}{d_j^{\text{out}}}$$

- Define $P = A^\top (D^{\text{out}})^{-1}$ where $D^{\text{out}} = \text{diag}[d_1^{\text{out}} \ \ldots \ d_n^{\text{out}}]$
- Define $\boldsymbol{r} = [r_1 \ \ldots r_n]^\top$
- **Observe:** $(P\boldsymbol{r})_i = \sum_j \frac{(A^\top)_{ij}}{d_j^{\text{out}}} r_j = \sum_j \frac{A_{ji}}{d_j^{\text{out}}} r_j = \sum_{j:\ (j,i)\in E} \frac{r_j}{d_j^{\text{out}}} = r_i$
- **Conclusion:** PageRank, if it exists, should satisfy $\boldsymbol{r} = P\boldsymbol{r}$.

Perron-Frobenius Theorem

- **Recall:** PageRank vector, $\boldsymbol{r}$, is 1-eigenvector of P.
- **Questions:**
  - Is 1 an eigenvalue of P?
  - Is there a unique eigenvector associated to 1?
  - We have implicitly assumed that $\boldsymbol{r} \geq 0$. Is this true?
- **Definition** a matrix A is called **reducible** if there exists permutation matrix Q such that:

$$QAQ^\top = \begin{bmatrix} X & Y \\ 0 & Z \end{bmatrix}$$

  otherwise A is **irreducible**.
- **Remark:** If A is symmetric, then it will be reducible if $QAQ^\top = \begin{bmatrix} X & 0 \\ 0 & Y \end{bmatrix}$.
- **Remark:** If A is adjacency matrix of directed graph G then A is irreducible $\Leftrightarrow$ G is strongly connected.
- **Theorem:** (Perron-Frobenius) Suppose that A is non-negative and irreducible:
  - $\lambda_1(A) = \rho(A)$, the spectral radius.
  - There is a unique eigenvector $\boldsymbol{v}_1$ corresponding to $\lambda_1(A)$.
  - $\boldsymbol{v}_1 > 0$

Perron-Frobenius Theorem

- $\rho(A) = \max\{|\lambda_1(A)|, \ldots, |\lambda_n(A)|\}$.
  - Note, assuming all eigenvalues are real, $\rho(A) = \pm\lambda_1(A)$.
- P is column-stochastic *i.e.* columns sum to 1.
- **Theorem:** Column-stochastic matrices have $\rho(P) = 1$.
- **Recall:** PageRank vector, if it exists, should satisfy $P\boldsymbol{r} = 1\boldsymbol{r} = \boldsymbol{r}$.
- Perron-Frobenius + Column-stochastic $\Rightarrow$ $\boldsymbol{r}$ exists and is unique.

On Markov Chains:

- Any column-stochastic P defines a Markov Chain via $\boldsymbol{v}^{t+1} = P\boldsymbol{v}^t$. Why?
- The $\boldsymbol{r}$ above is the **limiting/stationary distribution** of the Markov Chain:

$$\lim_{t\to\infty} P\boldsymbol{v}^t = \boldsymbol{r} \quad \text{for } \textit{any } \boldsymbol{v}^0$$

Teleportation

- For World Wide Web, P will very likely be reducible.
- **Solution:** add a small probability, $1 - \alpha$, of randomly jumping ("teleporting") from i to any other j:

$$\tilde{P} = \alpha P + (1 - \alpha)\mathbf{1}\mathbf{1}^\top$$

- **Check:** $\tilde{P}$ is still non-negative[1] and column stochastic.
- Previous theorem still applies: exists unique eigenvector $\boldsymbol{v}_1$ associated to eigenvalue 1.

[1] Actually, it is now positive, for which a stronger version of Perron-Frobenius applies

Computing v_1 using Power Method

- Have reduced ranking pages to computing leading eigenvector of $\tilde{P} = \alpha P + (1-\alpha)\mathbf{1}\mathbf{1}^\top \in \mathbb{R}^{n \times n}$.
- n may be very (very) large, but only need 1 (the leading) eigenvector. Can we do this efficiently?
- Suppose that $\tilde{P}$ has n distinct eigenvalues $\lambda_1, \ldots, \lambda_n$ with:

$$1 = \lambda_1 > |\lambda_2| > \ldots > |\lambda_n|$$

- *Any* $\boldsymbol{v} \in \mathbb{R}^n$ can be written as: $\boldsymbol{v} = \beta_1 \boldsymbol{v}_1 + \ldots + \beta_n \boldsymbol{v}_n = \sum_{i=1}^{n} \beta_i \boldsymbol{v}_i$
- $P\boldsymbol{v} = \sum_{i=1}^{n} \beta_i P \boldsymbol{v}_i = \sum_{i=1}^{n} \beta_i (\lambda_i \boldsymbol{v}_i) = \sum_{i=1}^{n} \beta_i \lambda_i \boldsymbol{v}_i = \beta_1 \boldsymbol{v}_1 + \sum_{i=2}^{n} \beta_i \lambda_i \boldsymbol{v}_i$
- Similarly: $P^2\boldsymbol{v} = \beta_1 \boldsymbol{v}_1 + \sum_{i=2}^{n} \beta_i \lambda_i^2 \boldsymbol{v}_i$
- More generally: $P^k\boldsymbol{v} = \beta_1 \boldsymbol{v}_1 + \sum_{i=2}^{n} \beta_i \lambda_i^k \boldsymbol{v}_i \to \beta_1 \boldsymbol{v}_1 + \sum_{i=2}^{n} \beta_i (0) \boldsymbol{v}_i = \beta_1 \boldsymbol{v}_1$

Computing $\boldsymbol{v}_1$ using Power Method

- **Power Method for Column Stochastic Matrix A**
  - Randomly initialize[2] $\boldsymbol{v}$
  - For $k = 1, \ldots, K$:
    1. $\boldsymbol{v} = P\boldsymbol{v}$
  - **Output: $\boldsymbol{v}$.**
- Argument on previous slide guarantees that $\boldsymbol{v} \approx \beta \boldsymbol{v}_1$, which is fine.

[2] Technically, we need to ensure that $\boldsymbol{v}^\top \boldsymbol{v}_1 \neq 0$, but randomly choosing coordinates of $\boldsymbol{v}$ will give this with probability 1

Summary

- PageRank is a procedure for assigning relevancy scores, r_i, to web pages on the same topic.
- $\boldsymbol{r}$ can be computed as the leading eigenvector of the matrix

$$\tilde{P} = \alpha A^\top (D^{\text{out}})^{-1} + (1-\alpha)\mathbf{1}\mathbf{1}^\top$$

- $\boldsymbol{r}$ can be computed efficiently using the Power Method.

Further Topics

- PageRank for other networks (gene regulatory, citation).
- PageRank for other applications (local clustering).
- Personalized PageRank.
- More efficient algorithms than Power Method.